

Raw Moving-Ball First Variation and Closure of the Bad-Density Step

Plan 2 / Level-Set Rearrangement Route

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Abstract

This note proves the remaining local theorem left by `faber_krahn_remaining_steps.tex`, with one correction: the right moving-ball vector field is not the singular unit field e_z , but the affine homothetic field $(x - z)/\rho + a$. It maps $B_\rho(z)$ exactly to $B_{\rho+h}(z + ha)$, is globally Lipschitz, and pairs with the existing homothetic radial-trace estimate. We prove a finite-perimeter one-sided first variation for

$$q(\rho) := \inf_{z \in \mathbb{R}^n} |E_\rho \Delta B_\rho(z)|.$$

The estimate controls only the positive derivative q'_+ , which is exactly what the endpoint trace needs. Together with nestedness, the bad-density set is paid at $O(\delta_T)$, and the bounded sharp Saint-Venant estimate follows from the level-set identity, same-centre Fusco–Julin, and the boundary-layer transfer.

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1 Setup

Let $\Omega \subset B_R$ be bounded, $|\Omega| = \omega_n$, and let u be the torsion potential. Set

$$E_\rho := \{u > t(\rho)\}, \quad |E_\rho| = \omega_n \rho^n, \quad w(\rho) := -t'(\rho), \quad d\mu = w(\rho) d\rho.$$

For a.e. ρ , E_ρ has finite perimeter, $|\nabla u| > 0$ on $\partial^* E_\rho$ up to $\mathcal{H}^{n-1}$ -null sets, and

$$V_\rho = \frac{w(\rho)}{|\nabla u|}, \quad \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = P(B_\rho).$$

Write

$$H_{z,\rho}(x) := \frac{(x - z) \cdot \nu_\rho(x)}{\rho}, \quad \bar{V}_\rho := \frac{P(B_\rho)}{P(E_\rho)}.$$

The exact level-set identity supplies

$$\int_{\rho_*}^{\rho_\delta} (D_H(t(\rho)) + D_I(t(\rho))) d\mu(\rho) \leq C(n, \rho_*) \delta_T(\Omega). \quad (1)$$

Talenti's profile-gap estimate supplies

$$|B_\tau| \leq C(n, \rho_*) \delta_T, \quad B_\tau := \{\rho : w(\rho) < \tau_0\}, \quad \tau_0 := \frac{\rho_*}{2n}. \quad (2)$$

Finally set

$$G_I := \{\rho : D_I(t(\rho)) \leq \eta_I\}.$$

On G_I , the same-centre Fusco–Julin package and the oriented radial trace lemma give, at a same-centre field z_ρ^{sc} ,

$$|E_\rho \Delta B_\rho(z_\rho^{\text{sc}})|^2 \leq C(n, R, \rho_*) D_I(t(\rho)), \quad (3)$$

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_*) D_I(t(\rho)), \quad (4)$$

where $z_\rho = z_\rho^{\text{sc}}$ in (4). The first inequality implies the same bound for $q(\rho)^2$, since q is an infimum over centres.

2 A Coarea Differentiation Lemma

The proof uses the following standard consequence of BV coarea and Lebesgue differentiation. It is the precise replacement for smooth normal-flow coordinates.

Lemma 2.1 (Variable-thickness coarea differentiation). *For a.e. regular level s of u , and every bounded Borel function $g \geq 0$,*

$$\lim_{h \downarrow 0} \frac{1}{h} |\{x : 0 < u(x) - s < hg(x)\}| = \int_{\partial^* \{u > s\}} \frac{g(x)}{|\nabla u(x)|} d\mathcal{H}^{n-1}(x), \quad (5)$$

and the analogous identity holds for $\{x : 0 < s - u(x) < hg(x)\}$.

Proof. For simple $g = \sum_{j=1}^N c_j \mathbf{1}_{A_j}$, the left side is a finite sum of terms

$$\frac{1}{h} \int_s^{s+hc_j} \int_{\partial^* \{u > \tau\}} \frac{\mathbf{1}_{A_j}(x)}{|\nabla u(x)|} d\mathcal{H}^{n-1}(x) d\tau,$$

by the BV coarea formula. At every common Lebesgue point of the finitely many coarea densities

$$\tau \mapsto \int_{\partial^* \{u > \tau\}} \frac{\mathbf{1}_{A_j}}{|\nabla u|} d\mathcal{H}^{n-1},$$

this converges to the right side of (5). Approximate a bounded Borel g from below and above by simple functions and use the monotonicity of the slabs. A countable monotone-class argument gives the identity for all bounded Borel g outside one null set of levels. The lower slab is identical. $\square$

3 Raw Moving-Ball First Variation

Lemma 3.1 (Affine shell estimate). *Fix a regular level ρ satisfying the differentiability and coarea Lebesgue-point properties above. Let $z, a \in \mathbb{R}^n$, and define*

$$T_h(x) := z + ha + \left(1 + \frac{h}{\rho}\right)(x - z).$$

Then, for $h \downarrow 0$,

$$\limsup_{h \downarrow 0} \frac{|E_{\rho+h} \Delta T_h(E_\rho)|}{h} \leq \int_{\partial^* E_\rho} |V_\rho - H_{z,\rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}. \quad (6)$$

Proof. The map $T_h = \text{id} + hW$ has the Lipschitz velocity

$$W(x) = \frac{x - z}{\rho} + a.$$

Since $u \in W_{\text{loc}}^{2,p}(\Omega)$ for $p > n$, hence $C_{\text{loc}}^{1,\alpha}$, a countable compact-exhaustion of $\{u = t(\rho)\} \cap \Omega$ gives, up to an $\mathcal{H}^{n-1}$ -null boundary part,

$$u(T_h^{-1}x) = u(x) - h \nabla u(x) \cdot W(x) + o(h)$$

uniformly on each compact patch. Also

$$t(\rho + h) = t(\rho) - hw(\rho) + o(h).$$

Therefore the symmetric difference is contained, modulo an $o(h)$ -thick slab, in the set where $u(x) - t(\rho)$ lies between 0 and

$$h(w(\rho) + \nabla u(x) \cdot W(x))$$

or between that number and 0. Applying Lemma 2.1 to the positive and negative parts gives

$$\limsup_{h \downarrow 0} \frac{|E_{\rho+h} \Delta T_h(E_\rho)|}{h} \leq \int_{\partial^* E_\rho} \frac{|w(\rho) + \nabla u \cdot W|}{|\nabla u|} d\mathcal{H}^{n-1}.$$

On $\partial^* E_\rho$, the outer normal is $\nu_\rho = -\nabla u / |\nabla u|$, so

$$\frac{w + \nabla u \cdot W}{|\nabla u|} = V_\rho - W \cdot \nu_\rho = V_\rho - H_{z,\rho} - a \cdot \nu_\rho.$$

This proves (6). □

Theorem 3.2 (One-sided first variation of the moving-ball distance). *For a.e. $\rho \in [\rho_*, \rho_\delta]$, let*

$$q(\rho) := \inf_z |E_\rho \Delta B_\rho(z)|.$$

Then

$$D^+ q(\rho) := \limsup_{h \downarrow 0} \frac{q(\rho + h) - q(\rho)}{h} \leq \frac{n}{\rho_*} q(\rho) + \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}, \quad (7)$$

where z_ρ is any minimiser of $z \mapsto |E_\rho \Delta B_\rho(z)|$.

Proof. The minimiser exists because $z \mapsto |E_\rho \Delta B_\rho(z)|$ is continuous and tends to $2|E_\rho|$ as $|z| \rightarrow \infty$. Fix such z_ρ , fix $a \in \mathbb{R}^n$, and use the affine map T_h from Lemma 3.1. Since

$$T_h(B_\rho(z_\rho)) = B_{\rho+h}(z_\rho + ha),$$

we have

$$\begin{aligned} q(\rho + h) &\leq |E_{\rho+h} \Delta T_h(B_\rho(z_\rho))| \\ &\leq |E_{\rho+h} \Delta T_h(E_\rho)| + |T_h(E_\rho) \Delta T_h(B_\rho(z_\rho))| \\ &= |E_{\rho+h} \Delta T_h(E_\rho)| + \left(1 + \frac{h}{\rho}\right)^n q(\rho). \end{aligned}$$

Subtracting $q(\rho)$, dividing by h , taking the limsup, and using $\rho \geq \rho_*$ and Lemma 3.1, gives

$$D^+ q(\rho) \leq \frac{n}{\rho} q(\rho) + \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}.$$

Take the infimum over a . □

Remark 3.3 (Why one-sided is enough). A two-sided derivative estimate would require controlling possible jumps of the optimal centre. The endpoint trace only transports information forward from a Markov radius to ρ_δ . Thus the positive derivative $(q')_+$, supplied by (7), is the exact quantity needed.

4 Integrated Positive Kinetic Bound

Lemma 4.1 (Transfer from the FJ centre to a distance-minimising centre). *Let $\rho \in G_I$. Let z_ρ^{sc} be a same-centre Fusco–Julin centre satisfying (3)–(4), and let z_ρ^{min} minimise $z \mapsto |E_\rho \Delta B_\rho(z)|$. Then*

$$\left(\int_{\partial^* E_\rho} |H_{z_\rho^{\text{min}}, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C(n, R, \rho_*) D_I(t(\rho)). \quad (8)$$

Proof. By minimality,

$$|E_\rho \Delta B_\rho(z_\rho^{\text{min}})| \leq |E_\rho \Delta B_\rho(z_\rho^{\text{sc}})| \leq C D_I(t(\rho))^{1/2}.$$

Hence

$$|B_\rho(z_\rho^{\text{min}}) \Delta B_\rho(z_\rho^{\text{sc}})| \leq |E_\rho \Delta B_\rho(z_\rho^{\text{min}})| + |E_\rho \Delta B_\rho(z_\rho^{\text{sc}})| \leq C D_I^{1/2}.$$

For two equal balls with centres separated by d , one has

$$|B_\rho(z_1) \Delta B_\rho(z_2)| \geq c(n) \rho^{n-1} \min\{d, \rho\}.$$

Choosing η_I small in the definition of G_I , this gives

$$|z_\rho^{\text{min}} - z_\rho^{\text{sc}}| \leq C(n, \rho_*) D_I(t(\rho))^{1/2}.$$

The homothetic trace is affine in the centre:

$$H_{z_\rho^{\text{min}}, \rho} - H_{z_\rho^{\text{sc}}, \rho} = -\frac{(z_\rho^{\text{min}} - z_\rho^{\text{sc}}) \cdot \nu_\rho}{\rho}.$$

On G_I , $P(E_\rho) \leq C(n, \rho_*, R)$. Therefore

$$\int_{\partial^* E_\rho} |H_{z_\rho^{\text{min}}, \rho} - H_{z_\rho^{\text{sc}}, \rho}| d\mathcal{H}^{n-1} \leq C(n, R, \rho_*) D_I(t(\rho))^{1/2}.$$

Combine this with (4) and square. □

Lemma 4.2 (Unconditional Lipschitz bound for q). *For $0 < \rho \leq \sigma \leq 1$,*

$$|q(\sigma) - q(\rho)| \leq 2\omega_n(\sigma^n - \rho^n) \leq 2n\omega_n|\sigma - \rho|.$$

Proof. Use the same centre for the two balls and the nestedness $E_\rho \subset E_\sigma$, $B_\rho(z) \subset B_\sigma(z)$:

$$q(\sigma) \leq q(\rho) + |E_\sigma \setminus E_\rho| + |B_\sigma \setminus B_\rho| = q(\rho) + 2\omega_n(\sigma^n - \rho^n),$$

and reverse the roles of ρ, σ . □

Proposition 4.3 (Positive kinetic estimate). *Let $J \subset [\rho_*, \rho_\delta]$ be an order-one interval. Then*

$$\int_J (q'_+(\rho))^2 d\rho \leq C(n, R, \rho_*, J) \delta_T(\Omega),$$

where $q'_+ = \max\{q', 0\}$, defined a.e. by the Lipschitz property.

Proof. Split J into

$$G := J \cap G_I \cap G_\tau, \quad \mathcal{B} := J \setminus G.$$

On G , Theorem 3.2, (3), and

$$\inf_a \int |V_\rho - H_{z_\rho^{\min}, \rho} - a \cdot \nu_\rho| \leq \int |V_\rho - \bar{V}_\rho| + \int |H_{z_\rho^{\min}, \rho} - \bar{V}_\rho|$$

give, after squaring,

$$(q'_+)^2 \leq C(n, R, \rho_*)(D_H(t(\rho)) + D_I(t(\rho))).$$

Here

$$\left(\int |V_\rho - \bar{V}_\rho| \right)^2 \leq D_H(t(\rho))$$

is the unconditional velocity-variance identity, and (4) controls the homothetic term after the transfer Lemma 4.1. Since $G \subset G_\tau$, $d\rho \leq \tau_0^{-1} d\mu$, so (1) gives

$$\int_G (q'_+)^2 d\rho \leq C\delta_T.$$

On $\mathcal{B}$, Lemma 4.2 gives $q'_+ \leq 2n\omega_n$. Also

$$|J \cap B_\tau| \leq C\delta_T$$

by (2), while

$$|J \cap G_\tau \cap G_I^c| \leq \tau_0^{-1} \mu(G_I^c) \leq \frac{C}{\eta_I} \int_J D_I d\mu \leq C\delta_T.$$

Thus $|\mathcal{B}| \leq C\delta_T$, and

$$\int_{\mathcal{B}} (q'_+)^2 d\rho \leq C\delta_T.$$

□

5 Endpoint Trace and Bounded Stability

Lemma 5.1 (Moving-ball endpoint trace). *Let $J = [\rho_\delta - L, \rho_\delta] \subset [\rho_*, \rho_\delta]$, with $L > 0$ fixed. Then*

$$q(\rho_\delta)^2 \leq C(n, R, \rho_*, L) \delta_T(\Omega).$$

Proof. First,

$$\int_{J \cap G} q(\rho)^2 d\rho \leq C \int_{J \cap G} D_I(t(\rho)) d\rho \leq C \int_J D_I(t(\rho)) d\mu(\rho) \leq C\delta_T.$$

On $J \setminus G$, the measure is $O(\delta_T)$ by the proof of Proposition 4.3, and $q \leq 2\omega_n$, so

$$\int_J q(\rho)^2 d\rho \leq C\delta_T.$$

Hence there is $\rho_0 \in J$ with $q(\rho_0)^2 \leq CL^{-1}\delta_T$. Since $\rho_0 \leq \rho_\delta$, the one-sided transport gives

$$q(\rho_\delta) \leq q(\rho_0) + \int_{\rho_0}^{\rho_\delta} q'_+(\rho) d\rho \leq C\delta_T^{1/2} + L^{1/2} \left(\int_J (q'_+)^2 d\rho \right)^{1/2} \leq C\delta_T^{1/2},$$

using Proposition 4.3. □

Theorem 5.2 (Bounded sharp Saint–Venant stability). *Assume the level-set identity, Talenti bad-density estimate, same-centre Fusco–Julin package, and finite-perimeter oriented radial trace lemma in the Plan 2 repository. Then for bounded $\Omega \subset B_R$,*

$$\mathcal{A}(\Omega)^2 \leq C(n, R) \delta_T(\Omega).$$

Proof. Choose

$$\rho_\delta = 1 - \kappa \sqrt{\delta_T(\Omega)}$$

with the usual fixed $\kappa = \kappa(n, R)$ and δ_T small enough that the order-one window $J = [\rho_\delta - L, \rho_\delta]$ lies in $[\rho_*, \rho_\delta]$. Lemma 5.1 gives

$$\mathcal{A}(E_{\rho_\delta})^2 \leq C(n, \rho_*) q(\rho_\delta)^2 \leq C\delta_T.$$

Since $E_{\rho_\delta} \subset \Omega$,

$$|\Omega \setminus E_{\rho_\delta}| = \omega_n(1 - \rho_\delta^n) \leq C(n) \sqrt{\delta_T}.$$

The elementary boundary-layer transfer gives

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\rho_\delta}) + \frac{2}{\omega_n} |\Omega \setminus E_{\rho_\delta}| \leq C \sqrt{\delta_T}.$$

Squaring proves the theorem. The large-deficit case is absorbed by reducing the constant, since $\mathcal{A}(\Omega) \leq 2$. □

6 What Changed

The remaining step is closed in the bounded route, but with two corrections to the earlier formulation.

1. The deformation field is homothetic:

$$W(x) = \frac{x - z}{\rho} + a,$$

not $e_z + a$. This avoids all singular-centre issues and uses exactly the homothetic trace already proved in Plan 2.

2. The first variation is one-sided:

$$D^+q \leq Cq + \text{homothetic boundary action}.$$

This is sufficient because the trace transports from a Markov radius forward to ρ_δ . A two-sided estimate is unnecessary.

3. Bad levels are harmless for q , because nestedness gives a genuine perimeter-free Lipschitz bound for the unscaled moving-ball distance. The invalid old claim concerned the rescaled sets $\rho^{-1}E_\rho$, not q .

All constants are computable from the constants in the level-set identity, Talenti profile gap, same-centre Fusco–Julin, the radial trace lemma, and the explicit Lipschitz constants above.

References

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